

Gun Violence in America: Using Random Forests to Predict Gun Death Intent

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1. Introduction / Background

In recent history, gun violence has emerged as one of the most pressing social issues in the United States. Motivated by interest in this topic, a research team consisting of Matthew Reardon (Statistics major, DiDa minor), Miguel Gonzalez (Actuarial major, Economics major), and Ryan Smith (Actuarial major, DiDa minor) analyzed data pertaining to gun-related deaths. These deaths typically fall into one of four main classifications or causes of intent: suicides, homicides, accidental, and undetermined.

Our team had access to data provided by the CDC that can be found at this link:

https://raw.githubusercontent.com/fivethirtyeight/guns-data/refs/heads/master/full_data.csv. The data consists of 100,798 entries with eleven different columns. The columns consist of an unnamed index column, and the other ten features are: year, month, intent, police, sex, age, race, hispanic, place, and education. The primary goal of the research is to create models that are able to accurately predict the cause of gun death (listed as “intent” in the data) based on the other demographic and situation variables that the data provides.

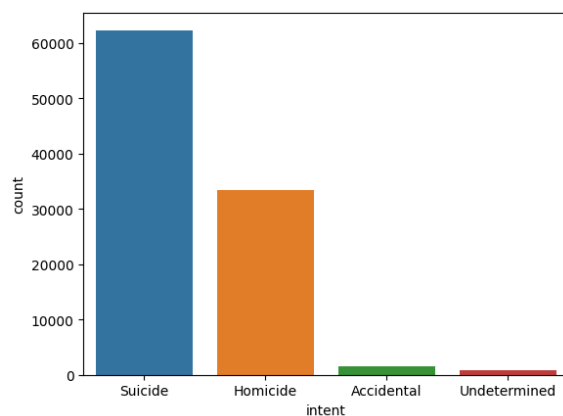
During the data cleaning process, the “Unnamed : 0” index column was dropped as well as the “hispanic” column as it didn’t affect the model and we were unsure what the ambiguous values represented. Random forest models were generated showing the relationship between gun death intent and the explanatory variables, allowing our team to determine which features were most important when classifying the different forms of intent. Random Forest

models were favored as our data set is large, unbalanced, and nonlinear, and the models were robust to noise.

2. Explanatory Data Analysis

2.1. Intent Overview

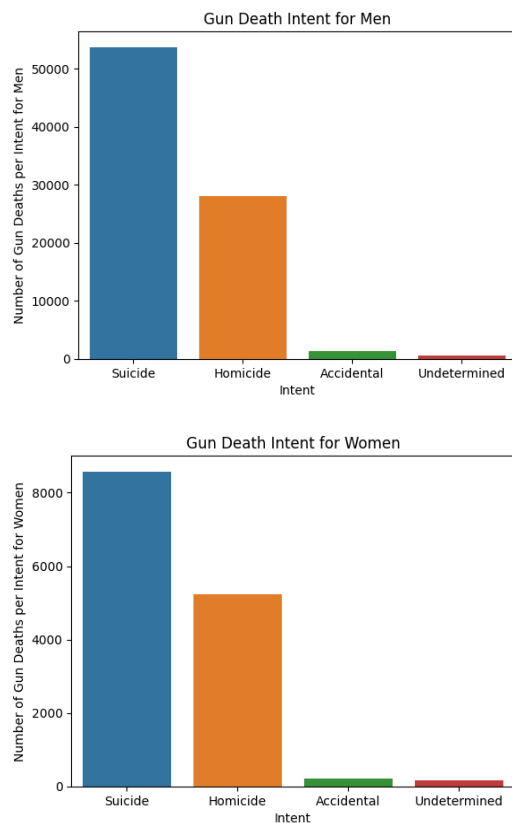
Before developing the model, we performed exploratory data analysis (EDA) to gain a deeper understanding of the data. This allowed us to see how different variables related to gun death intent. The EDA also allowed us to detect unbalanced data and determine which features had the greatest predictive power.



The figure shows that suicides and homicides make up the majority of gun deaths with suicide being the leading cause. However, it’s also clear that accidental and undetermined intent are very unrepresentative, as they only make up approximately 2% of the data.

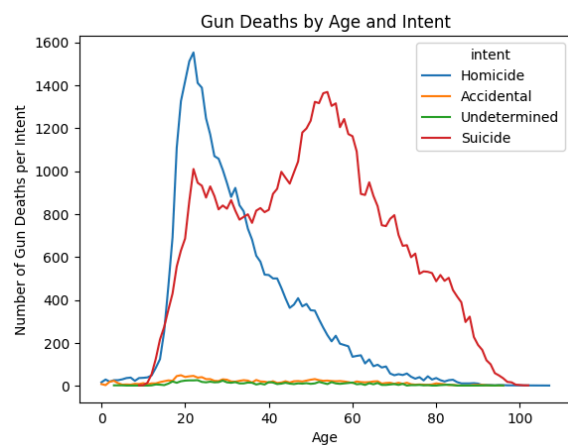
We proceed to look at the features: sex, age, place, race, and education to investigate any outlying trends in the data.

2.2. Sex



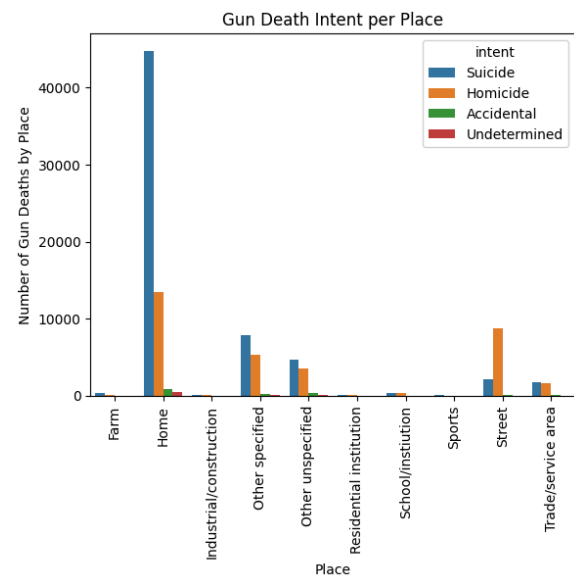
The count of gun deaths is significantly lower for women. However, the proportions of intent appear to show the same trend as before with the majority of deaths being suicides and homicides and accidental and undetermined deaths being very unrepresentative in the data.

2.3. Age



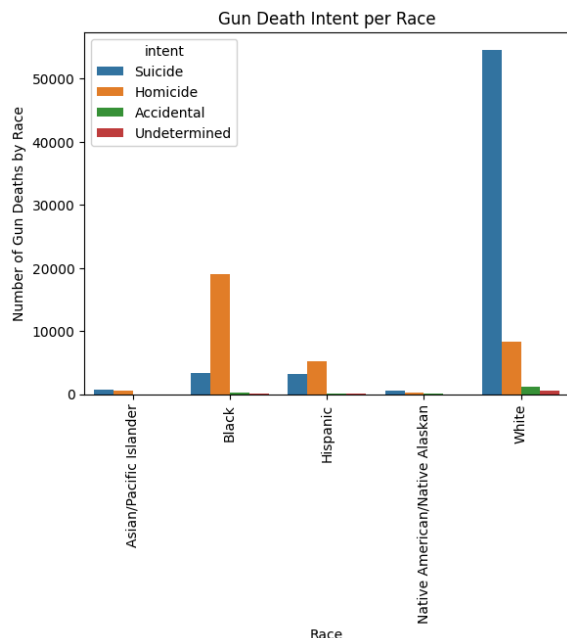
For younger victims (around twenty years old), homicide was much more likely to be the intent compared to suicide. On the other hand, for victims aged 50-60, suicide makes up a much higher proportion of gun death intent. Unlike sex, age seems to be a much more important feature when predicting intent as there are stark differences in proportions of intent between different age groups.

2.4. Place



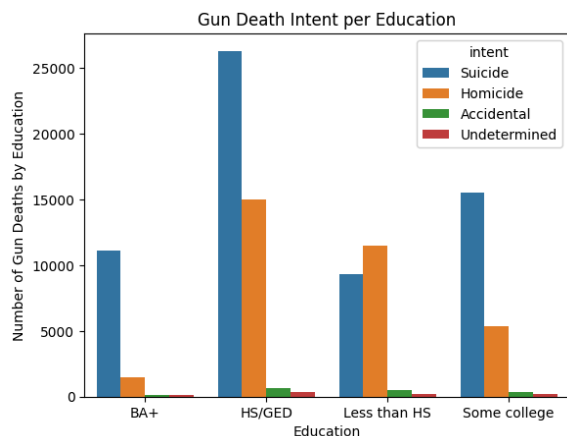
The figure shows that most deaths occur at home and of those deaths, the majority are suicides. In almost every location, suicide is the most common form of intent, although, in the street, homicide is the leading cause of death. The clear difference between intent at home vs the street leads us to view these locations as features that could potentially be important.

2.5. Race



The analysis on intent vs race gives the most data for blacks, hispanics, and whites. For both black and hispanic victims, the most common intent was homicide, with black homicide making up a bigger proportion of black intent compared to hispanic homicide vs hispanic intent. White victims had the highest number of gun deaths and the vast majority came from suicide. These three races, specifically black and white seem to differ greatly in their relation to intent and could be features with high predictive importance.

2.6. Education

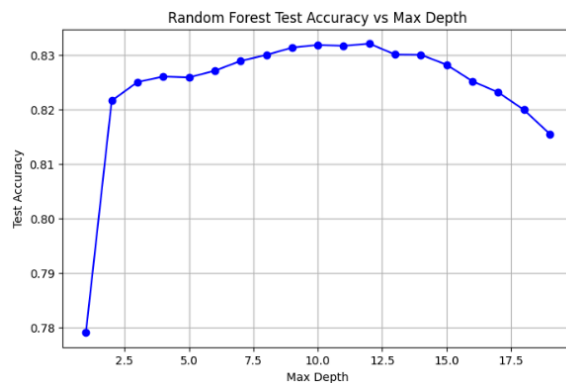


Suicide is the most common cause of death for all levels of education apart from "Less than HS" with homicide leading as the most common form of intent. The count of suicides for those with an education level of HS/GED is much higher compared to all other levels of education. This leads us to believe this feature could also have high importance in our model.

3. Predictive Modeling

3.1. Multiclass Random Forest

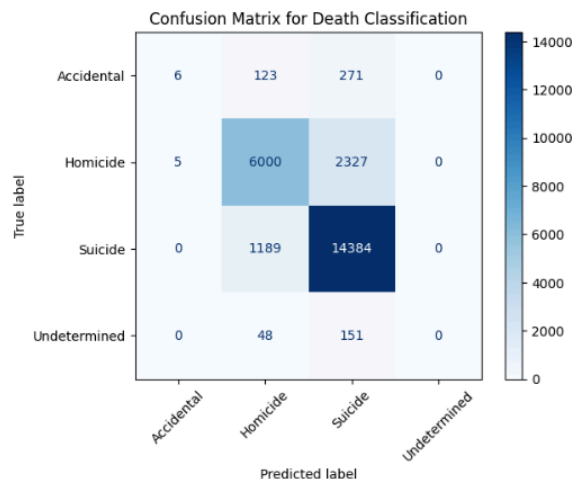
To extend our analysis, we built predictive models to determine whether demographic and contextual variables could accurately predict the intent behind a gun death. We started by implementing Random Forest classifiers to capture the interactions between predictors age, race, sex, education, place of death, police. We first trained a multiclass Random Forest model predicting all four intents. We split the data into a training set and test set with 75% and 25% of the data respectively and stratification to keep class proportions intact. We then used dummy variables for race, sex, education, and place so the model could run. We tested maximum depths from 1-20.



We saw with no max depth, the test accuracy was 79%. After running the model we saw that test accuracy was maximized at 83% with a max depth of 12. Our model is properly fit with a training accuracy of 85% and a test accuracy of 83%. Our model is also 19% better than the null accuracy suggesting that our model is significantly better than randomly guessing. We

then ran a k-fold cross validation with 5 folds. We got the mean validation score to be 83% with a standard deviation of 0.7%. This indicates that model accuracy is consistent among all folds and generalizes well to unseen data. We also ran a classification report and confusion matrix to further analyze the strength of our model.

	precision	recall	f1-score	support
Accidental	0.55	0.01	0.03	400
Homicide	0.82	0.72	0.76	8332
Suicide	0.84	0.92	0.88	15573
Undetermined	0.00	0.00	0.00	199
accuracy			0.83	24504
macro avg	0.55	0.41	0.42	24504
weighted avg	0.82	0.83	0.82	24504

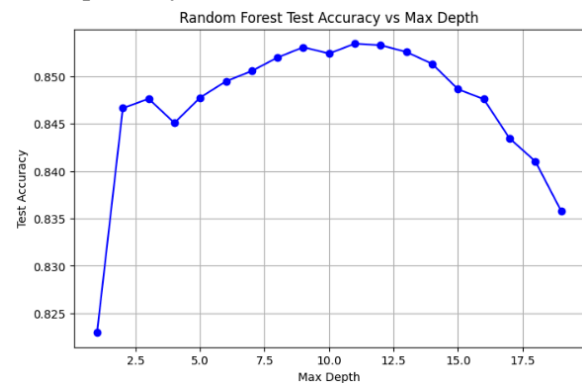


The model predicts Suicide and Homicide very well but rarely predicts Accidental and Undetermined correctly. In our confusion matrix, you can see that Undetermined was never predicted correctly. We can see our macro average is 55%, which indicates our model is poor among all classes. This is because Accidental only makes up 1.6% of the data and Undetermined makes up only .8% of the data. We found the PR-AUC per class and found consistent results with classification reports. The PR-AUC for Accidental, Homicide, Suicide, and Undetermined were 6%, 84%, 92%, and 1% respectively. The homicide and suicidal PR-AUC were very good, while accidental and undetermined were very poor, but again it is expected as they are very underrepresented. The macro and micro averages just prove this, as they are 46% and 88% respectively. We also had an ROC-AUC of 76% which is pretty good, but

is being dragged down by accident and undetermined.

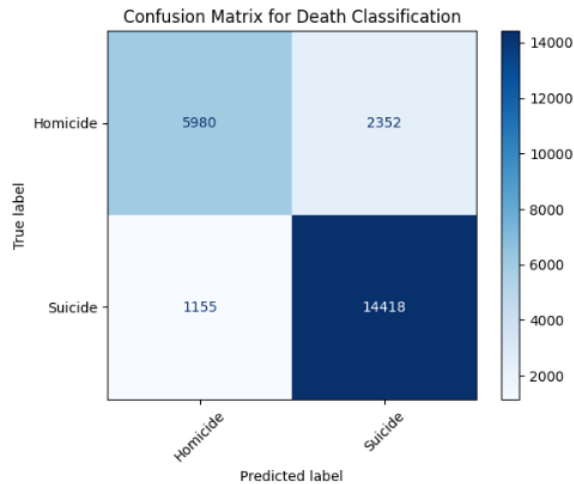
3.2. Binary Classification Model

Due to Accidental and Undetermined amount to less than 3% of the data, lower PR-AUC scores, and a poor macro average, we created a second model trained to only predict Homicide versus Suicide cases. This would most likely create a better model, as both the classes it's trying to predict have good representation. So, we created a binary Random Forest Model with the reduced dataset containing 95,620 rows with predictors age, race, sex, education, place of death, police, year, and month.



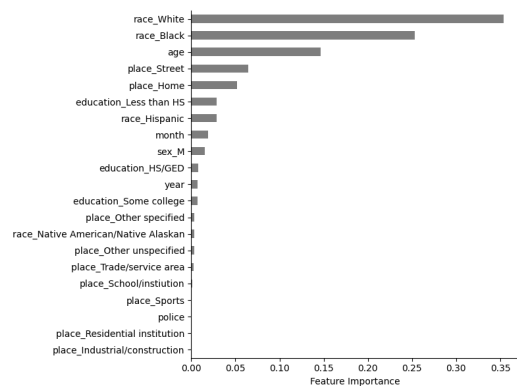
This time our best test accuracy was 85% at a max depth of 11. Our new model's test accuracy is 20% better than null accuracy, slightly better than before. This model also cross-validated well, yielding a mean score of 85% and standard deviation of 0.7%, again indicating that the model is consistent among all folds and generalities to unseen data very well. The classification report and confusion matrix also yield slightly better results.

	precision	recall	f1-score	support
Homicide	0.84	0.72	0.77	8332
Suicide	0.86	0.93	0.89	15573
accuracy			0.85	23905
macro avg	0.85	0.82	0.83	23905
weighted avg	0.85	0.85	0.85	23905



The new model predicts Suicide and Homicide slightly better. We can most notably see that our macro average increased from 55% to 85%. It has a better PR-AUC and ROC-AUC than the previous model, as they are 88% and 92% respectively. Due to its significant improvements in ROC-AUC, PR-AUC, and macro average, we chose the binary classification model to be our main research model.

Now that we are more confident in our binary model, we can start to interpret our model to find the most important predictors.



We can see that races white and black and age were most important when making predictions in this model. Whether it happens in the street or home also seems to have importance. This makes sense, as our EDA goes along with these results.

When looking at these influential features, it's important to remember that these patterns reflect broader social conditions rather than individual characteristics. For example, the difference between races likely occurs from systemic issues in the conditions of different communities. These differences could include exposure to violence or economic opportunity. Similarly, the differences observed amongst the locations of crime relate to well known differences in social risk of suicides and homicides. The data may also have some limitations such as reporting biases, demographic labels not capturing the full identities of individuals, and possible misclassification. Our model is useful for identifying population-level trends but shouldn't be used to make assumptions about individuals as this could reinforce inequity. The possible biases and different context of cases as well as the implications from the use of demographic information should be carefully considered when using the model.

4. Conclusion

Our predictive modeling demonstrates that Suicidal and Homicidal gun-death intent can be predicted with strong accuracy using demographics, age, environment, and other predictors. These models reinforce the idea that firearm deaths follow demographic and situational patterns rather than at random. Using this knowledge can help us further discuss the social, economic, and environmental factors underlying gun violence; however, it is important to recognize that predictive data alone does not capture the full story of human behavior. Broader societal context, lived experiences and structural conditions must also be considered when interpreting these results.